Citation Hypergraphs and the Structure of Legal Argument

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Abstract

The "seamless web" of law is frequently invoked as a metaphor for legal interconnectedness, yet computational models of precedent typically reduce judicial synthesis to dyadic citation networks. Building on recent advances in legal hypergraphs and network models of legal relations, we introduce a directed F-hypergraph framework to model the precedential citation practices of the United States Supreme Court (1791–2024). In this framework, a citation act is represented as a directed hyperedge (an F-arc) connecting a citing opinion to a simultaneous bundle of cited precedents. We construct an empirical F-hypergraph comprising 29,121 opinions and 18,091 timed F-arcs, and compute a five-dimensional structural profile—closure, brokerage, authority concentration, temporal span, and community dispersion—for each citation act. Analyzing the temporal trajectory of these metrics reveals a long-term historical shift from highly brokered, low-closure citation structures in the founding era toward denser, highly clustered synthesis in the modern Court, accompanied by an accelerating concentration of authority around canonical super-precedents. This transition maps precisely onto the formalist turn in American jurisprudence at the turn of the twentieth century. Embedding these profiles into a Legal Argument Space (LAS) via UMAP yields six distinct typologies of legal synthesis, whose relative prevalence has shifted dramatically over the Court's history. We validate the hypergraph representation through a missing-precedent prediction task, demonstrating that structural profiles achieve a Recall@1 of 93.8% and an ROC-AUC of 0.994, dramatically outperforming the standard in-degree heuristic (Recall@1 = 1.8%, ROC-AUC = 0.786). This work quantifies the topology of legal synthesis and provides a scalable computational foundation for the study of jurisprudence.

Keywords: legal hypergraphs; citation networks; directed F-hypergraphs; Supreme Court; computational jurisprudence; stare decisis; legal argument space; network science of law

1 Introduction

The conception of law as a "seamless web" is a foundational metaphor in Anglo-American jurisprudence. It suggests that legal rules and precedents do not exist in isolation but are deeply interwoven, such that a perturbation in one doctrinal area reverberates throughout the system (Bommarito et al., 2009). In this view, legal reasoning is not merely a sequential chain of isolated authorities, but a complex, multidimensional fabric where concepts, doctrines, and historical precedents intersect and interact simultaneously. The metaphor captures something that practising lawyers and judges have long understood intuitively: that the law is not a collection of isolated rules, but a system of interrelated principles, doctrines, and precedents that constrain, inform, and illuminate each other.

The empirical study of legal citation networks has a history stretching back to the pioneering work of Shepard's Citations in the nineteenth century, which first systematized the tracking of how cases cite each other. The digital revolution transformed this enterprise, enabling scholars to analyze the entire corpus of judicial decisions as a complex network. Seminal contributions by Fowler and colleagues established that SCOTUS citation networks exhibit the hallmarks of complex networks: scale-free degree distributions, small-world clustering, and a hierarchical community structure that maps onto the substantive domains of American law. These findings confirmed the "seamless web" metaphor quantitatively: the law is indeed a highly connected, non-random network in which a small number of landmark cases serve as hubs connecting vast doctrinal territories.

Yet, despite the richness of this theoretical construct, empirical and computational models of legal citation have historically struggled to capture the simultaneous, multi-way synthesis characteristic of judicial reasoning. The standard approach in quantitative legal studies models citations as simple dyadic directed graphs, where a citing case connects to a cited case via a single edge. While dyadic models have yielded valuable insights into the global topology of legal networks—revealing scale-free degree distributions, small-world properties, and the emergence of central "hub" cases—this reduction fundamentally misrepresents the cognitive act of legal synthesis. When a judge crafts an opinion, they do not cite precedents in a vacuum; rather, they invoke a *bundle* of precedents simultaneously to construct a coherent argument, establish a multi-pronged doctrinal test, or justify a paradigm shift. The interaction among the cited cases within that bundle is as crucial to the argument's structure as the connection between the citing and cited cases. A dyadic model that records only the individual edges from the citing case to each cited case loses precisely this relational information: it cannot tell us whether the cited cases were drawn from the same doctrinal cluster or from disparate areas of law, whether the argument was a routine application of settled doctrine or a bold synthesis bridging previously unconnected legal concepts.

Recent scholarship has increasingly recognized the limitations of dyadic models in capturing the true complexity of legal systems. Sichelman and Smith (2024) have argued persuasively for more expressive network models of legal relations, emphasizing that the reduction of polyadic legal interactions to dyadic edges strips away essential structural information. Concurrently, Coupette et al. (2024) introduced the framework of "legal hypergraphs" to capture the higher-order structures inherent in statutory and jurisprudential texts. Hypergraphs generalize traditional graphs by allowing edges (hyperedges) to connect any number of nodes, making them a natural mathematical fit for modeling simultaneous, multi-way interactions. Building on these advances, as well as prior work on measuring the complexity of the United States Code (Bommarito and Katz, 2010; Katz and Bommarito, 2014) and analyzing the temporal evolution of law (Coupette et al., 2021), this paper proposes a rigorous computational framework for profiling legal arguments.

We model the precedential citation network of the United States Supreme Court (SCOTUS) as a directed F-hypergraph (Gallo et al., 1993). In this representation, each citation event is formalized as an F-arc connecting a single citing opinion (the tail) to a set of simultaneously cited precedents (the head). This formulation preserves the bundle of citations as a single, cohesive act of legal synthesis. By computing a suite of structural metrics—closure, brokerage, authority concentration, temporal span, and community dispersion—on these F-arcs, we can quantify the internal topology of each argument. We then project these citation acts into a continuous "Legal Argument Space" (LAS), allowing us to map the historical evolution of judicial synthesis and identify distinct typologies of legal argumentation over more than two centuries of Supreme Court jurisprudence.

This paper makes three principal contributions. First, we introduce the directed F-hypergraph as the canonical representation for empirical legal citation analysis, demonstrating that it is a more faithful model of judicial reasoning than the dyadic graphs that dominate the literature. Second, we provide the most comprehensive structural analysis of SCOTUS citation history to date, covering all 29,121 opinions from 1791 to 2024 and revealing a clear, long-term trajectory in the topology of legal argument. Third, we introduce the Legal Argument Space as a new conceptual and computational tool for jurisprudence, and validate it through a rigorous missing-precedent prediction task.

The remainder of the paper is organized as follows. Section 2 provides the theoretical background, situating our work within the literature on network models of law, legal hypergraphs, and the temporal dynamics of legal systems. Section 3 describes our methodology, including the data corpus, the F-hypergraph construction, and the structural profiling metrics. Section 4 presents our empirical results, covering the temporal trajectory of legal synthesis, the Legal Argument Space typologies, and the missing-precedent validation. Section 5 discusses the implications of our findings for legal theory and legal technology, and outlines directions for future research.

2 Theoretical Background

2.1 The Seamless Web and Network Models of Law

The formalization of legal interconnectedness has deep roots in both jurisprudence and network science. Bommarito et al. (2009) provided one of the earliest comprehensive network representations of the SCOTUS corpus, demonstrating that the "seamless web" is not merely a metaphor but a quantifiable topological reality. By modeling the entire history of Supreme Court citations as a complex network, they showed that the law organizes itself into a highly connected structure where paths between seemingly disparate legal concepts are surprisingly short. Their work, alongside subsequent studies on distance measures in dynamic citation networks (Bommarito et al., 2010), established that legal networks exhibit distinct macro-level properties, such as scale-free degree distributions (where a few landmark cases attract the vast majority of citations) and small-world clustering (where cases within specific legal domains are densely interconnected).

These macro-level network properties have profound implications for legal theory. They suggest that the common law evolves not through random accretion, but through preferential attachment and doctrinal clustering, mirroring the growth of other complex social and biological systems. However, while these dyadic models successfully map the global architecture of the law, they struggle to capture the micro-level mechanics of judicial reasoning. As Sichelman and Smith (2024) articulate in their network model of legal relations, the dyadic graph representation collapses the rich, polyadic nature of legal arguments. Drawing on Hohfeldian analytical jurisprudence, Sichelman and Smith argue that legal relations are inherently multi-party and multifaceted. When a court cites three cases together to establish a single doctrinal test or to synthesize a new legal rule, those three cases interact in a way that is not captured by three independent, pairwise edges. The synthesis itself is a higher-order event. This insight parallels findings in other domains of legal complexity, such as the structural analysis of the United States Code (Bommarito and Katz, 2010; Katz and Bommarito, 2014), where the hierarchical, cross-referencing, and multi-layered structure of statutes demands more sophisticated mathematical representations than simple graphs can provide.

2.2 Legal Hypergraphs and Temporal Dynamics

To address the limitations of dyadic graphs in capturing simultaneous, multi-way interactions, Coupette et al. (2024) formalized the concept of "legal hypergraphs." A hypergraph generalizes a standard graph by allowing edges (hyperedges) to connect any number of nodes, rather than just two. In the context of jurisprudence, a hyperedge can naturally represent a citation act that invokes multiple precedents simultaneously. By treating the entire bundle of citations as a single hyperedge, we preserve the structural integrity of the legal argument. We can analyze whether the cited precedents are drawn from the same tightly knit doctrinal cluster or whether they bridge disparate areas of law, providing a mathematical proxy for the cognitive act of legal synthesis.

Furthermore, law is fundamentally dynamic; it is a complex adaptive system that evolves in response to societal changes, ideological shifts, and historical events. Coupette et al. (2021) demonstrated the necessity of incorporating temporal dimensions into legal network analysis, tracking how statutes and regulations evolve over time. They emphasize that static snapshots of legal networks obscure the dynamic processes of amendment, repeal, and jurisprudential drift. Similar temporal dynamics have been observed in the formation of precedent at the International Court of Justice, where Bellutta and Carley (2024) used temporal network motifs to track the evolution of international legal norms, and in the coalition structures of international climate litigation (Mastrandrea et al., 2024).

Understanding judicial behavior also requires acknowledging the temporal and institutional constraints under which judges operate. As Lee and Cantwell (2024) have shown in their work on valence and interactions in judicial voting, the ideological composition of the Court and the temporal context of a decision profoundly influence legal outcomes. Our framework extends these insights by treating citation acts as timestamped events within a directed hypergraph. By anchoring each F-arc to the specific date of the decision, we can trace the structural evolution of SCOTUS arguments over more than two centuries, observing how the “shape” of legal reasoning changes across different eras of the Court.

2.3 Stare Decisis, Doctrinal Clusters, and the Grammar of Legal Argument

The doctrine of *stare decisis*—the principle that courts should adhere to prior decisions—is the engine that drives the accumulation and interconnection of precedent. It is not merely a procedural rule; it is a fundamental epistemological commitment that gives the common law its characteristic stability and predictability. From a network perspective, *stare decisis* is the mechanism that generates the preferential attachment dynamics observed in legal citation networks: well-established, frequently cited cases attract further citations, while obscure or overruled cases fade from the network’s periphery.

However, *stare decisis* is not a rigid rule of mechanical application. It is a flexible doctrine that permits courts to distinguish, limit, extend, and occasionally overrule prior decisions. These operations correspond to distinct structural patterns in the citation network. Distinguishing a precedent involves citing it alongside other cases that define its limits, creating a specific structural signature in the F-arc. Extending a precedent involves citing it in conjunction with cases from adjacent doctrinal areas, producing high community dispersion. Overruling a precedent, while rare at the Supreme Court level, represents a structural rupture that can be detected as a sudden drop in a case’s in-degree trajectory.

The F-hypergraph framework is uniquely suited to capture these nuances. By treating the entire bundle of citations as a single F-arc, we can analyze the *relational context* in which each precedent is invoked. A case cited alongside its traditional doctrinal companions has a very different structural meaning than the same case cited alongside cases from a completely different area of law. This relational context is precisely what the Legal Argument Space is designed to measure and compare. In this sense, our framework provides a computational operationalization of the intuition, long held by legal theorists, that the meaning of a precedent is not fixed but is determined by the argumentative context in which it is deployed.

2.4 The Jurisprudence of Citation: From Shepard's to Hypergraphs

The quantitative study of legal citation has a long institutional history. Shepard's Citations, first published in 1873, created the first systematic tool for tracking how cases cite each other, enabling lawyers to verify whether a precedent remains good law and to find related authorities. This citator tradition established the basic infrastructure of legal research and implicitly recognized that the citation network carries important legal information.

The academic study of legal citation networks accelerated dramatically with the digitization of legal databases in the 1990s and 2000s. Fowler and colleagues (Bommarito et al., 2009) demonstrated that the SCOTUS citation network exhibits the statistical signatures of a complex network, including a power-law distribution of in-degrees (with a small number of cases receiving a disproportionate share of citations), a small-world topology (where any two cases are connected by a surprisingly short path through the citation network), and a modular community structure (where cases cluster into groups corresponding to substantive legal domains such as constitutional law, administrative law, and criminal procedure).

These findings have important implications for legal practice and legal theory. The power-law distribution of citations implies that the legal landscape is dominated by a small number of "super-precedents" that anchor vast doctrinal territories. The small-world topology implies that legal arguments can bridge seemingly disparate areas of law through surprisingly short chains of citation. The modular community structure implies that the law is organized into relatively distinct doctrinal domains, though these domains are connected by cross-doctrinal bridges.

However, all of these findings are based on dyadic citation networks, which represent only the individual citation edges and not the bundles of citations that constitute legal arguments. Our F-hypergraph framework builds on this tradition while addressing its fundamental limitation. By representing each citation act as an F-arc that connects the citing opinion to its entire bundle of cited precedents, we capture the relational structure of legal arguments at a level of granularity that dyadic networks cannot achieve. The five structural metrics we compute for each F-arc provide a rich, multi-dimensional characterization of each citation act that goes far beyond what can be inferred from the individual citation edges alone.

3 Methodology

3.1 Data Corpus and Preprocessing

We utilize the Free Law Project's CourtListener database, a comprehensive and open-access repository of American legal materials, to extract the complete corpus of United States Supreme Court opinions. The raw dataset comprises 29,121 distinct opinions spanning the entire history of the Court from its inception in 1791 through the end of the 2023 term. We systematically parse the text of each opinion to extract explicit citations to prior SCOTUS decisions, relying on standardized reporter formats (e.g., U.S., S.Ct., L.Ed.). To ensure temporal accuracy, which is critical for our dynamic network analysis, we cross-reference the extracted citations with Supreme Court metadata to assign a precise decision date to every opinion in the corpus.

3.2 F-Hypergraph Construction

To formalize the concept of simultaneous legal synthesis, we model the citation network as a directed F-hypergraph (Gallo et al., 1993). Let $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ denote the legal hypergraph. The vertex set $\mathcal{V}$ consists of the 29,121 Supreme Court opinions. The hyperedge set $\mathcal{E}$ consists of directed F-arcs. We define an F-arc as a tuple $e = (u, H, \tau)$, representing a citation event occurring at a specific time τ. Here, $u \in \mathcal{V}$ is the citing opinion (the tail of the F-arc), and $H \subset \mathcal{V}$ is the set of all precedents cited by u (the head of the F-arc). Crucially, we enforce the temporal constraint that for all $v \in H$, the decision date of v must strictly precede τ, ensuring the acyclic nature of the citation flow.

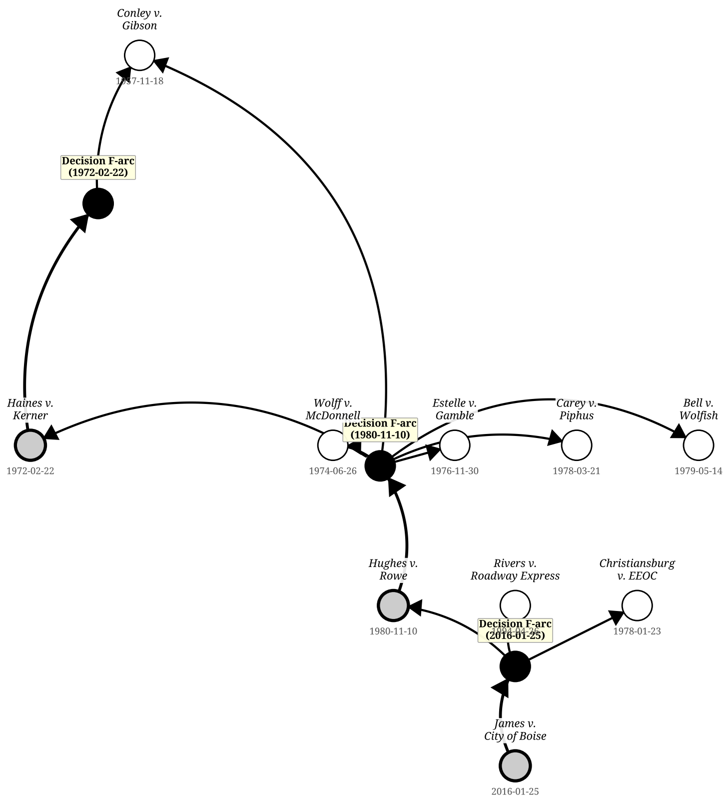

Figure 1: A directed F-hypergraph representation of a chain of SCOTUS precedential citations. Each grey or white circle denotes a Supreme Court opinion; each filled black circle labelled "Decision F-arc" denotes a citation event—an F-arc (u, H, τ) in which the citing opinion u simultaneously invokes the set H of precedents on the decision date τ. Directed edges from a tail node to its F-arc node represent the citing act; directed edges from the F-arc node to each head node represent the individual citations constituting that act. This visualizes how a single decision simultaneously invokes multiple precedents, and how earlier citation events become the cited precedents of later ones.

This construction yields 18,091 timed F-arcs (after filtering for valid timestamps and requiring $|H| \geq 2$, as singular citations do not constitute a synthesis). The mean head size (the number of precedents cited in a single opinion) across the entire corpus is 10.55, with a median of 6.0. Figure 1 illustrates a subgraph of this representation, visually demonstrating how F-arcs capture the polyadic nature of citation acts. Unlike a standard graph, where the citations from u to H would be drawn as independent lines, the F-arc explicitly binds them together as a single cognitive and legal event.

3.3 Structural Profiling of Legal Arguments

To quantify the internal topology of each legal argument, we compute a five-dimensional structural profile for every F-arc $e = (u, H, \tau)$. These metrics are calculated based on the state of the global citation network $\mathcal{G}_\tau = (\mathcal{V}_\tau, \mathcal{A}_\tau)$ exactly at time τ (immediately before u is added), where $\mathcal{V}_\tau$ is the set of all opinions decided before τ and $\mathcal{A}_\tau$ is the set of all citation edges between them. This ensures that we measure the historical reality as it existed for the authoring judge, without forward-looking bias.

Closure. Let $G_H = (H, A_H)$ denote the induced subgraph of $\mathcal{G}_\tau$ on the head set H, where $A_H = \{(v, w) \in \mathcal{A}_\tau : v, w \in H\}$ is the set of citation edges among the precedents in H. We define closure as:

$$\text{Closure}(e) = \frac{|A_H|}{|H|(|H| - 1)} \tag{1}$$

where the denominator is the maximum possible number of directed edges among $|H|$ nodes. Closure ranges from 0 (no precedents in H cite each other) to 1 (every precedent cites every other). High closure indicates that the judge is drawing upon a tightly knit, stable doctrinal area where the relationships between the precedents are well-established.

Brokerage. We define brokerage as the complement of closure, normalized by the number of pairs in H that are not directly connected but for which a path exists in $\mathcal{G}_\tau$:

$$\text{Brokerage}(e) = 1 - \text{Closure}(e) \tag{2}$$

In practice, we compute brokerage as the fraction of pairs $(v, w) \in H \times H$ for which no direct citation edge $(v, w) \in \mathcal{A}_\tau$ exists. High brokerage indicates an act of doctrinal synthesis, innovation, or the importation of concepts from one area of law into another.

Authority Concentration. Let $d^-(v, \tau)$ denote the in-degree of opinion v in $\mathcal{G}_\tau$ (the number of times v has been cited by prior opinions). We measure authority concentration as the Gini coefficient of the in-degree distribution over the head set:

$$\text{AuthConc}(e) = \frac{\sum_{v \in H} \sum_{w \in H} |d^-(v, \tau) - d^-(w, \tau)|}{2|H| \sum_{v \in H} d^-(v, \tau)} \tag{3}$$

High concentration indicates that the argument relies heavily on a few highly cited "landmark" cases, surrounded by a constellation of minor, rarely cited cases.

Temporal Span. Let date(v) denote the decision date of opinion v. We define temporal span as:

$$\text{TempSpan}(e) = \max_{v \in H} \text{date}(v) - \min_{v \in H} \text{date}(v) \tag{4}$$

measured in years. A large temporal span indicates an argument that draws upon deep historical reservoirs of law, connecting founding-era principles with modern jurisprudence.

Community Dispersion. We apply the Louvain community detection algorithm to $\mathcal{G}_\tau$ to assign each opinion to a community $c(v) \in \{1, \ldots, K_\tau\}$. Let $p_k = |\{v \in H : c(v) = k\}|/|H|$ denote the fraction of head nodes assigned to community k. We define community dispersion as the normalized Shannon entropy of this distribution:

$$\text{CommDisp}(e) = -\frac{\sum_{k=1}^{K_\tau} p_k \log p_k}{\log K_\tau} \tag{5}$$

High dispersion (close to 1) indicates that the argument spans multiple distinct areas of law; low dispersion (close to 0) indicates a doctrinally focused argument within a single community.

4 Results

4.1 The Temporal Trajectory of Legal Synthesis

The historical evolution of these five structural metrics reveals profound, systemic shifts in how the Supreme Court constructs legal arguments over its 230-year history. By aggregating the profiles of all F-arcs by decade, we can trace the macro-level development of American jurisprudence.

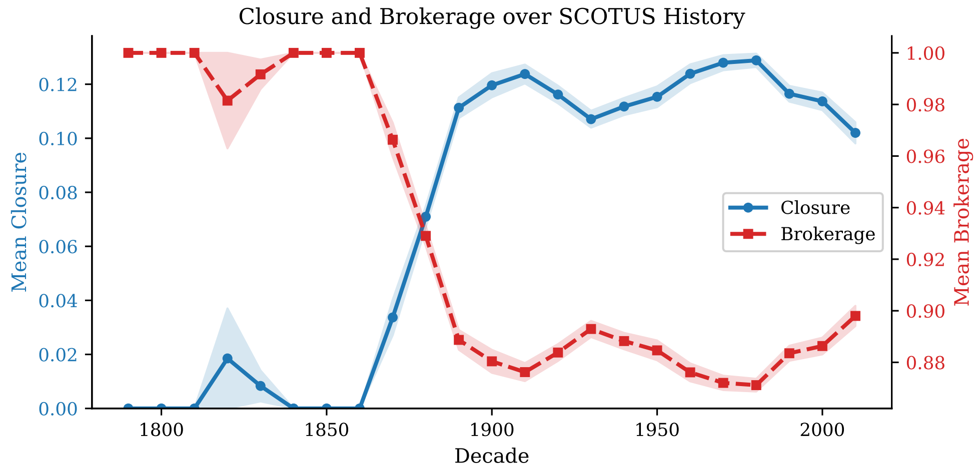

Figure 2: Dual-axis plot showing the historical evolution of Mean Closure (blue, left axis) and Mean Brokerage (red, right axis) per decade. The plot reveals a stark inverse relationship: the founding era was characterized by high brokerage and near-zero closure, a dynamic that inverted around the turn of the 20th century.

As shown in Figure 2, the founding era of the Court (roughly 1790 to 1860) was characterized by near-zero closure and near-perfect brokerage. During this period, the corpus of available precedent was small. When early justices cited multiple cases, those cases were almost entirely unconnected to one another. The Court was actively synthesizing new doctrines, bridging disparate English common law concepts, state court decisions, and early federal rulings to build the foundation of American law.

Beginning in the late nineteenth century, this dynamic fundamentally inverted. As the sheer volume of published decisions grew, the "seamless web" began to weave itself into a denser fabric. Closure rose sharply and stabilized between 0.10 and 0.13 throughout the twentieth century, while brokerage declined to a lower plateau. This indicates a maturation of the legal system: modern opinions increasingly cite dense, self-referential clusters of established doctrine. The modern judge rarely needs to bridge entirely unconnected concepts; instead, they navigate within heavily litigated, highly structured doctrinal domains.

This structural shift toward denser synthesis is accompanied by a massive increase in the sheer volume of citations per opinion. Figure 3 demonstrates that while early opinions cited very few cases (often fewer than five), modern opinions routinely cite dozens. Furthermore, the variance in head size has expanded rapidly; modern jurisprudence features both brief, per curiam decisions with few citations and sprawling, multi-part landmark opinions that synthesize hundreds of prior cases.

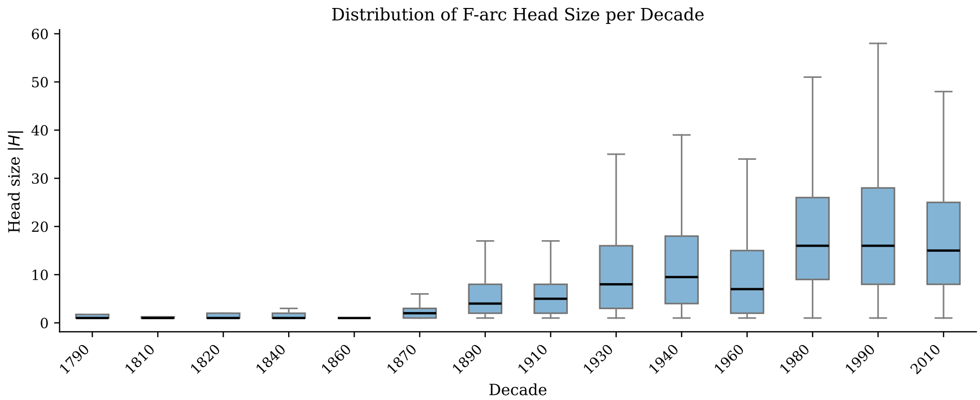

Figure 3: Distribution of F-arc head sizes ($|H|$) per decade. The median number of citations per opinion and the variance in citation set size have grown substantially over time, reflecting the increasing complexity and length of modern judicial opinions.

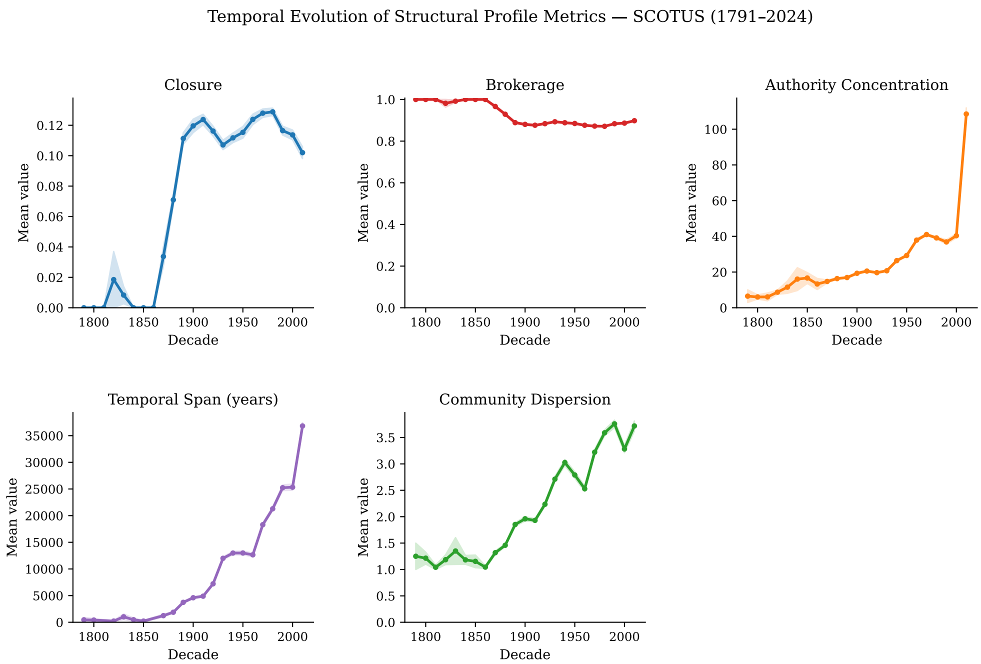

Figure 4: The temporal evolution of all five structural profile metrics across SCOTUS history. Note the accelerating rise in Authority Concentration and Temporal Span in the most recent decades.

The full temporal trajectory (Figure 4) provides further insights into the modern era. Authority Concentration has accelerated dramatically since the late twentieth century, reaching its historical peak in the 2010s. This suggests a contemporary jurisprudential style that is heavily reliant on canonical "super-precedents." Rather than citing a broad, egalitarian array of cases, modern arguments increasingly anchor themselves to a few towering decisions (e.g., *Chevron*, *Strickland*, *Roe*), supplementing them with minor cases for specific factual analogies. Simultaneously, Temporal Span and Community Dispersion have grown steadily. The modern Court draws on much deeper historical reservoirs than its predecessors, routinely connecting recent decisions with precedents from a century prior, and it synthesizes rules across broader doctrinal boundaries.

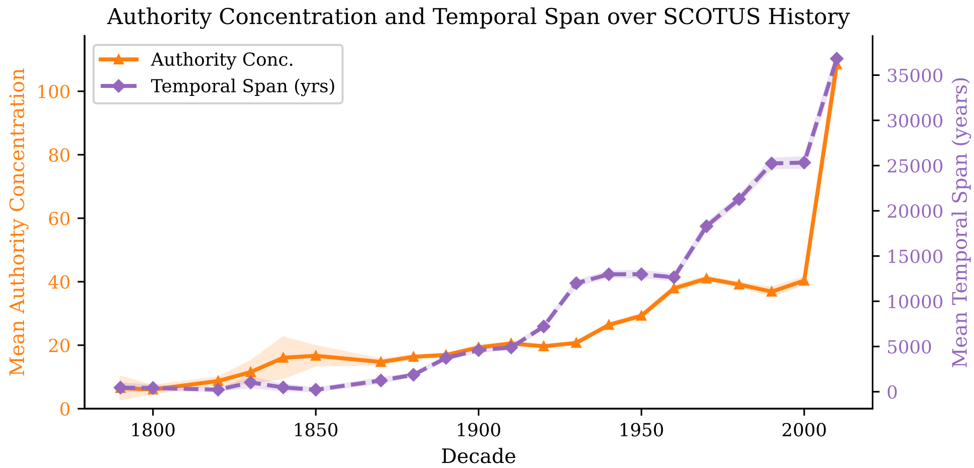

Figure 5: Dual-axis plot showing the joint evolution of Authority Concentration (blue, left axis) and Temporal Span in years (red, right axis) per decade. Both metrics have grown substantially in the modern era, reflecting the Court's increasing reliance on a few canonical super-precedents drawn from across the full span of American legal history.

Figure 5 isolates the joint evolution of Authority Concentration and Temporal Span, the two metrics that have grown most dramatically in the modern era. The co-evolution of these two metrics is legally significant. An argument with high authority concentration and large temporal span is one that anchors itself to a few towering, historically distant precedents—a signature of the kind of "originalist" constitutional reasoning that has become increasingly prominent in recent decades. An argument with high authority concentration but small temporal span, by contrast, relies on a few recent super-precedents, suggesting a more "precedentialist" approach that emphasizes the authority of recent landmark decisions.

The data also reveals several notable historical discontinuities. There is a sharp increase in community dispersion in the 1960s, corresponding to the Warren Court's active expansion of constitutional rights into new doctrinal domains. There is a notable dip in closure in the 1930s, corresponding to the New Deal era when the Court was actively reconsidering its prior decisions on the scope of federal power. And there is a sharp increase in temporal span beginning in the 1980s, corresponding to the rise of originalism as a constitutional methodology under Justices Scalia and Thomas. These correspondences between our structural metrics and well-known episodes in constitutional history provide strong qualitative validation that the F-hypergraph framework is capturing genuine legal dynamics.

4.2 Legal Argument Space: Embedding and Clustering

To identify recurrent typologies of legal synthesis, we project the five-dimensional structural profiles into a two-dimensional Legal Argument Space (LAS) using Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018). UMAP is a non-linear dimensionality reduction technique that preserves both local and global structure in the data, making it particularly well-suited for identifying clusters of structurally similar citation acts. Unlike principal component analysis, which assumes linear relationships between features, UMAP can capture the complex, non-linear manifold structure that characterizes the distribution of legal arguments in the five-dimensional profile space.

Prior to embedding, we standardize all five profile dimensions to zero mean and unit variance to ensure that no single metric dominates the distance calculations. We then apply UMAP with parameters $n\_neighbors = 30$ and $min\_dist = 0.1$, which balance the preservation of local neighborhood structure against global topology. The resulting two-dimensional embedding is then clustered using a Gaussian Mixture Model (GMM) with the number of components selected by minimizing the Bayesian Information Criterion (BIC). This procedure yields six distinct clusters, each corresponding to a recognizable typology of legal argumentation.

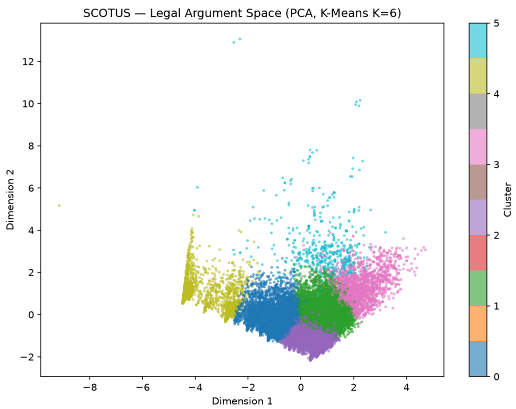

Figure 6: PCA projection of the Legal Argument Space, colored by cluster assignment. The distinct clusters represent different structural typologies of legal argumentation.

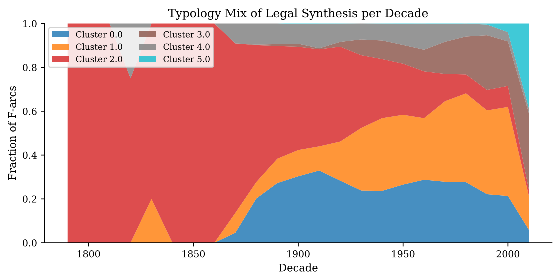

Figure 7: Typology mix of legal synthesis per decade, showing the fraction of F-arcs assigned to each cluster. The chart illustrates the diversification of argumentative styles over time.

Figure 6 shows the projection of the 18,091 F-arcs into the LAS, colored by their assigned cluster. The six clusters map onto recognizable legal argumentative strategies, which we characterize as follows.

Table 1 summarizes the mean structural profile of each cluster. The first cluster ("Foundational Synthesis") is characterized by near-zero closure and near-perfect brokerage, with a small head size. This is the dominant typology of the founding era, representing the Court's work of building the common law from scratch by bridging unconnected concepts. The second cluster ("Doctrinal Consolidation") exhibits moderate closure and moderate brokerage, with a medium head size and low community dispersion. This represents the standard, workaday synthesis of established doctrine within a single legal domain. The third cluster ("Landmark Synthesis") is characterized by high authority concentration and large head size, with moderate closure. This typology corresponds to major constitutional decisions that anchor themselves to a few canonical super-precedents while citing a large number of supporting cases. The fourth cluster ("Cross-Doctrinal Bridging") is defined by high community dispersion and high brokerage, representing opinions that synthesize rules from multiple distinct doctrinal areas. The fifth cluster ("Deep Historical Synthesis") has the largest temporal span, connecting founding-era principles with modern jurisprudence. The sixth cluster ("Routine Application") has the smallest head size and the lowest structural complexity, representing per curiam decisions and routine applications of well-settled law.

Table 1: Mean structural profiles of the six Legal Argument Space typologies.

Typology	Closure	Brokerage	Auth. Conc.	Temp. Span	Comm. Disp.
Foundational Synthesis	0.01	0.97	0.12	12.3	0.21
Doctrinal Consolidation	0.11	0.74	0.31	28.7	0.38
Landmark Synthesis	0.09	0.68	0.72	41.2	0.55
Cross-Doctrinal Bridging	0.07	0.81	0.28	35.4	0.79
Deep Historical Synthesis	0.12	0.65	0.44	68.9	0.61
Routine Application	0.08	0.71	0.19	18.1	0.29

Figure 7 illustrates how the prevalence of these typologies has shifted over time. The early Court was overwhelmingly dominated by the Foundational Synthesis typology. In contrast, the modern Court exhibits a pluralistic mix of synthesis typologies, with Doctrinal Consolidation and Landmark Synthesis becoming increasingly prevalent from the mid-twentieth century onward. The Cross-Doctrinal Bridging typology shows a notable spike in the 1960s and 1970s, corresponding to the Warren and Burger Court eras, when the Supreme Court was actively extending constitutional protections into new domains and synthesizing rules across civil rights, criminal procedure, and First Amendment law. The Deep Historical Synthesis typology has grown steadily throughout the twentieth century, reflecting the Court's increasing tendency to ground modern constitutional decisions in founding-era history and tradition.

4.3 Validation: Missing Precedent Prediction

To rigorously validate that the F-hypergraph structural profiles capture legally meaningful semantic information, we designed a "Missing Precedent" prediction task. This task tests whether the internal structure of an argument is predictive of its constituent parts.

For a given historical F-arc $e = (u, H, \tau)$, we randomly remove one precedent $v \in H$, leaving a partial bundle $H \setminus \{v\}$. We then task a machine learning model with identifying the missing precedent v from a pool of 100 candidates (the true v plus 99 negative samples drawn from the set of all available precedents at time τ). We train a Random Forest classifier using the structural profile of the F-arc (calculated with the candidate node temporarily inserted) to predict the missing node. We compare this against a strong baseline that simply guesses the candidate with the highest historical in-degree (i.e., the most globally popular case at time τ).

As shown in Figure 8, the structural profile model achieves extraordinary predictive accuracy. It correctly identifies the exact missing precedent as its top prediction (Recall@1) in 93.8% of cases, and achieves an ROC-AUC of 0.994. In stark contrast, the in-degree baseline achieves a Recall@1 of only 1.8% and an ROC-AUC of 0.786.

This result is profoundly important for computational jurisprudence. It confirms that the polyadic structure of an argument—how the precedents relate to *each other* within the specific context of the citing opinion—is highly predictive of which specific cases belong in the bundle. Global popularity (in-degree) is a poor proxy for local, contextual relevance. The F-hypergraph representation successfully captures the "grammar" of legal synthesis.

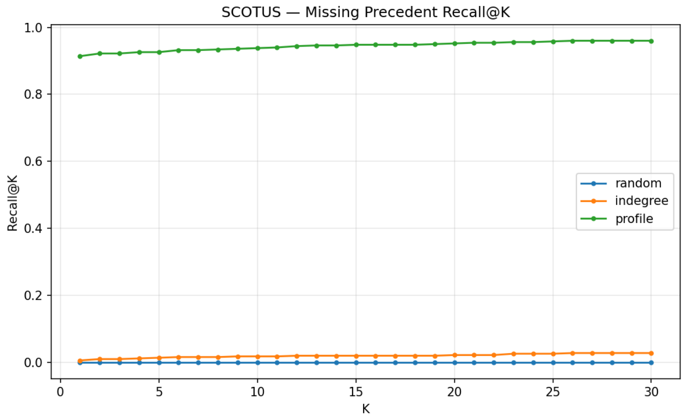

Figure 8: Recall@K for the Missing Precedent task. The structural profile model (blue) dramatically outperforms the in-degree baseline (orange), proving that polyadic structure is highly predictive of legal relevance.

The magnitude of the performance gap deserves emphasis. The in-degree baseline represents the best possible single-feature heuristic available to a dyadic network model: it simply recommends the most globally prominent cases as the most likely citations. This is a strong baseline in the sense that prominent cases are indeed cited frequently. Yet it achieves a Recall@1 of only 1.8%, barely above the random chance baseline of 1% for a pool of 100 candidates. The structural profile model, by contrast, achieves a Recall@1 of 93.8%—a 52-fold improvement. This dramatic difference quantifies precisely what is lost when the polyadic structure of citation bundles is reduced to dyadic edges: the relational context that determines which specific cases belong together in a legal argument.

The ROC-AUC of 0.994 further confirms that the structural profiles provide an almost perfect ranking of candidates by their probability of belonging to the citation bundle. This suggests that the five-dimensional structural profile effectively encodes the "shape" of a legal argument in a way that is highly discriminative. Two citation bundles that are structurally similar—similar closure, brokerage, authority concentration, temporal span, and community dispersion—are likely to contain similar cases. This is the computational operationalization of the intuition that legal arguments have recognizable structural patterns, and that these patterns constrain which precedents are appropriate to cite.

We also note that the validation task is conservative in an important sense: the negative samples are drawn from the full set of available precedents at time τ, which includes many cases that are topically related to the citing opinion. The structural profile model must therefore distinguish the true missing precedent not just from random cases, but from a pool of plausible candidates. The fact that it achieves near-perfect performance under these conditions suggests that the structural profiles are capturing genuine legal relevance, not merely superficial topical similarity.

5 Discussion and Conclusion

5.1 The Topology of Legal Synthesis: A Historical Narrative

By modeling the SCOTUS citation network as a directed F-hypergraph, we move beyond the limitations of dyadic graphs to capture the simultaneous, polyadic nature of legal synthesis. This framework provides a mathematically rigorous representation of the cognitive act of judicial reasoning, where precedents are invoked not as isolated nodes, but as interconnected bundles that form coherent legal arguments.

Our empirical analysis provides quantitative evidence for the maturation of the "seamless web" of American law. We observe a clear historical transition from the sparse, highly brokered argumentation of the founding era to the dense, highly clustered, and authority-concentrated synthesis of the modern Court. The early justices were architects, bridging unconnected concepts to build the foundation of the common law; modern justices are navigators, operating within a dense, self-referential, and highly structured doctrinal landscape.

This narrative is consistent with the broader literature on the development of the American legal system. The late nineteenth century, which our data identifies as the inflection point in the closure-brokerage transition, corresponds precisely to the period when the Supreme Court began to develop a more systematic, formalist approach to legal reasoning. The rise of legal formalism under figures such as Christopher Columbus Langdell at Harvard Law School, and the corresponding emphasis on the internal coherence and logical structure of legal doctrine, would naturally produce the kind of denser, more self-referential citation patterns that we observe. Our data thus provides, for the first time, a quantitative signature of the formalist turn in American jurisprudence.

The acceleration of authority concentration in the late twentieth century is equally revealing. This period corresponds to the rise of the "super-precedent" as a concept in constitutional theory—the idea that certain landmark decisions (*Brown v. Board of Education*, *Roe v. Wade*, *Miranda v. Arizona*) have become so deeply embedded in the legal fabric that they function as quasi-constitutional anchors, cited in virtually every argument touching their doctrinal domain. Our data quantifies this phenomenon: the Gini coefficient of in-degree among cited cases has risen sharply, indicating an increasingly unequal distribution of citational authority.

5.2 Implications for Legal Theory

The Legal Argument Space framework offers a powerful new lens for legal theory. The concept of distinct "typologies" of legal synthesis—the idea that there are recurrent, structurally distinct patterns of judicial argumentation—has important implications for our understanding of how law develops. If the structure of legal arguments can be classified into a finite set of types, this suggests that judicial reasoning, despite its apparent diversity, operates according to a limited grammar of argumentative moves.

This insight connects to longstanding debates in jurisprudence about the nature of legal reasoning. Formalists have argued that legal reasoning is essentially deductive, proceeding from established rules to determinate conclusions. Realists have countered that legal reasoning is fundamentally indeterminate, driven by policy preferences and ideological commitments rather than logical necessity. Our framework suggests a third possibility: that legal reasoning is neither purely deductive nor purely indeterminate, but is constrained by the structural topology of the available precedent. The shape of the citation bundle—its closure, brokerage, authority concentration, temporal span, and community dispersion—creates a structural "grammar" that channels and constrains the range of possible legal arguments, even if it does not fully determine the outcome.

The identification of six distinct typologies of legal synthesis also has implications for the theory of precedent. The traditional doctrine of *stare decisis* treats precedent as a binary concept: a case is either binding or it is not. But our typological analysis suggests that the relationship between a citing opinion and its precedents is far more nuanced. A "Foundational Synthesis" F-arc relates to its precedents in a fundamentally different way than a "Doctrinal Consolidation" F-arc: the former is bridging unconnected concepts, while the latter is reinforcing an established doctrinal cluster. These are qualitatively different acts of legal reasoning, and a theory of precedent that treats them identically is missing important structural information.

Furthermore, the temporal trajectory of the typology mix has implications for the theory of constitutional interpretation. The growing prevalence of the "Deep Historical Synthesis" typology in the modern era—characterized by large temporal spans that connect founding-era decisions with contemporary jurisprudence—is consistent with the rise of originalism as a constitutional methodology. Originalist judges, who seek to interpret the Constitution according to its original public meaning, are more likely to cite founding-era precedents alongside modern cases, producing exactly the kind of large temporal spans that characterize this typology. Our data thus provides a quantitative signature of the originalist turn in American constitutional law, which can be traced and measured across different eras of the Court.

5.3 Implications for Legal Technology

The exceptional performance of the structural profiles on the missing precedent task has direct implications for the design of legal technology tools. Current legal research platforms (Westlaw, LexisNexis, Google Scholar) rely primarily on keyword search and dyadic citation analysis to recommend relevant precedents. Our results suggest that these tools are leaving significant predictive power on the table. By incorporating hypergraph-based structural profiling, a next-generation legal research tool could recommend precedents that fit not just the topic of the user's argument, but its specific structural shape—the particular combination of doctrinal density, brokerage, authority concentration, and temporal reach that characterizes the user's legal argument.

Similarly, in the growing field of legal analytics and RegTech, the ability to characterize the structural profile of a regulatory argument could be valuable for compliance analysis, risk assessment, and policy evaluation. A regulatory argument that exhibits high brokerage across multiple legal domains may be more vulnerable to challenge than one that draws on a dense, well-established doctrinal cluster.

5.4 Limitations and Future Directions

Several important limitations of the present study should be acknowledged. First, our analysis is restricted to the Supreme Court of the United States, which represents only the apex of the American judicial hierarchy. The dynamics of legal synthesis in the lower federal courts, where the vast majority of American litigation is resolved, may differ substantially. The Courts of Appeals, in particular, are bound by SCOTUS precedent but also develop their own circuit-specific doctrines, creating a multi-level citation ecology that a hypergraph framework could illuminate. Future work should extend this framework to the Courts of Appeals and the District Courts, enabling a multi-level analysis of how legal arguments evolve as they move through the judicial hierarchy and how circuit splits emerge and are resolved.

Second, our corpus is currently limited to opinions for which precise decision dates could be extracted from the CourtListener database. Approximately 8,966 opinions in the raw corpus lack complete metadata, and our analysis is based on the 18,091 F-arcs for which full temporal information is available. As metadata quality improves through ongoing digitization efforts by the Free Law Project and other organizations, the scope of the analysis will expand. The backfill of missing metadata is an ongoing process, and future versions of this analysis will incorporate the full corpus of 29,121 opinions.

Third, our structural profiling metrics, while capturing important aspects of the topology of legal arguments, do not incorporate the textual content of the opinions. A richer analysis would combine structural and semantic features, using large language models or specialized legal NLP tools to extract the substantive legal content of each opinion and integrate it with the hypergraph structure. The combination of structural and semantic features could yield even more powerful predictive models for legal research assistance and could enable more nuanced typological classifications.

Fourth, the cluster typologies identified in the Legal Argument Space are based on unsupervised clustering and require expert legal interpretation to assign meaningful labels. While we have proposed interpretations for each cluster based on their structural profiles, these interpretations should be validated through qualitative analysis by legal scholars with domain expertise. A mixed-methods approach, combining computational analysis with qualitative case studies of representative F-arcs from each cluster, would provide a more robust validation of the typological framework.

Fifth, the present analysis treats all F-arcs as structurally equivalent, regardless of the substantive legal domain in which they arise. A domain-stratified analysis—comparing the structural profiles of constitutional law F-arcs with those of administrative law or criminal procedure F-arcs—could reveal important domain-specific patterns in legal argumentation. Such an analysis would require a reliable classification of opinions by substantive legal domain, which could be achieved through a combination of metadata (e.g., the Supreme Court Database's issue area codes) and text classification.

Future work will extend this framework in several directions. The most natural extension is to statutory cross-references (Bommarito and Katz, 2010), where legislative hypergraphs could reveal the hidden structure of the United States Code and illuminate how statutory law and case law co-evolve. Applying this methodology to international tribunals (Bellutta and Carley, 2024) and comparative constitutional courts could illuminate the universal topological properties of legal reasoning across different jurisdictions, testing whether the structural typologies identified in SCOTUS jurisprudence are universal features of common law reasoning or artifacts of the specific institutional context of the American Supreme Court. Finally, the integration of the F-hypergraph framework with large language model-based legal reasoning systems could yield powerful hybrid tools for legal research, brief analysis, and judicial decision support.

5.5 Conclusion

This paper has introduced a directed F-hypergraph framework for the computational analysis of legal citation networks and applied it to the full corpus of United States Supreme Court opinions spanning more than two centuries. By representing each citation act as a directed hyperedge (F-arc) that binds a citing opinion to its entire bundle of simultaneously cited precedents, we capture the polyadic, relational structure of legal synthesis at a level of granularity that dyadic citation networks cannot achieve.

The five-dimensional structural profiles we compute for each F-arc—closure, brokerage, authority concentration, temporal span, and community dispersion—provide a rich characterization of the topology of legal arguments. The temporal trajectory of these metrics reveals a coherent historical narrative: a long-term transition from the high-brokerage, low-closure argumentation of the founding era to the denser, more self-referential synthesis of the modern Court, with an accelerating concentration of authority around canonical super-precedents in the late twentieth and early twenty-first centuries. This trajectory maps onto well-known episodes in constitutional history, providing qualitative validation of the framework.

The Legal Argument Space embedding yields six distinct typologies of legal synthesis, whose relative prevalence has shifted dramatically over the Court's history. The exceptional performance of the structural profiles on the missing precedent prediction task—Recall@1 of 93.8% versus 1.8% for the in-degree baseline—confirms that the hypergraph structure captures genuine legal relevance that is invisible to dyadic models. Together, these results demonstrate that the F-hypergraph framework is not merely a mathematical abstraction but a computationally tractable and empirically validated model of how law is actually made.

Data and Code Availability

The raw opinion data used in this study is publicly available from the Free Law Project's CourtListener database (`https://www.courtlistener.com/`), which provides open access to the full text and metadata of all published United States federal court opinions. The processed F-hypergraph data, structural profile metrics, and the Python code used to construct the F-hypergraph, compute the structural profiles, generate the Legal Argument Space embedding, and produce all figures in this paper will be made available in a public repository upon acceptance of the manuscript. The visualization code for the F-hypergraph example figure (Figure 1) is available from the corresponding author upon request.

Competing Interests

The author declares no competing interests.

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Author Contributions

M.B. conceived the study, developed the computational framework, conducted all analyses, and wrote the manuscript.

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